

Course Title

BITS Pilani

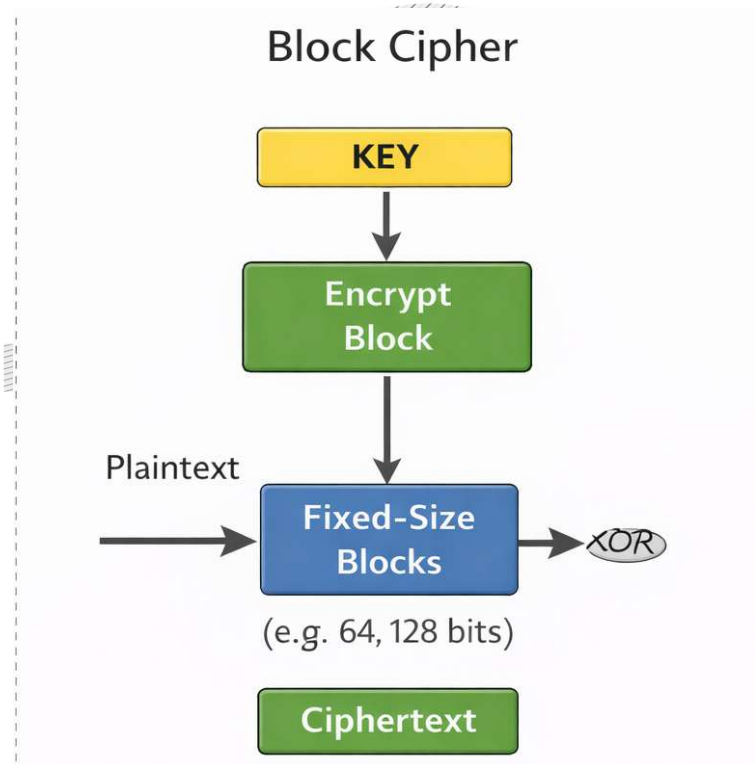
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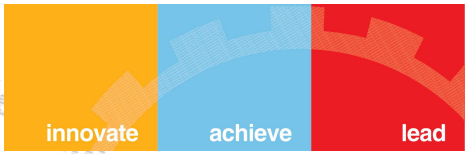
ES ZG513/ SS ZG513, Network Security
Lecture No IV.

Block Cipher

Block cipher: Encrypts data in fixed-size blocks (like 64-bit or 128-bit) using a key-based transformation. Large messages are encrypted by processing many blocks one after another.



Shannon Principles in Block Cipher Design



Claude Shannon proposed two fundamental ideas for secure cipher design:

1. **Confusion**
2. **Diffusion**

Confusion:

The relationship between the **secret key** and the **ciphertext** should be highly complex and hidden. An attacker should not be able to predict how changing the key affects the output.

Diffusion:

Each plaintext bit influences **many ciphertext bits**.
Small input change ⇒ large unpredictable output change.

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Goal of Confusion

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1. Hide statistical structure.
2. Prevent algebraic or linear relations.
3. Make key recovery difficult.

How Confusion is Achieved
Main tool: Non-linear Substitution (S-Boxes)

Why nonlinear?

1. Linear functions are easy to analyze.
2. Nonlinear mappings destroy patterns

Example

$C = P \oplus K$ Without confusion Attacker easily relates key and ciphertext.
$C = S(P \oplus K)$ With confusion the relation becomes complex.

Goal of Diffusion

1. Spread information across the block
2. Remove statistical patterns
3. Achieve avalanche effect

How Diffusion is Achieved
Common techniques:
1. Permutations
2. Linear mixing (matrix operations)
3. Bit shuffling
4. MixColumns (AES)

Before diffusion: 10000000
After diffusion: 10110110

“Diffusion spreads the data.”

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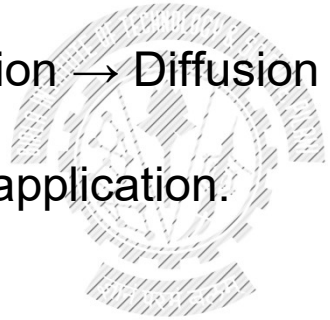
How Modern Ciphers Use Both

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Every round of AES or DES combines:

Confusion → Diffusion → Confusion → Diffusion ...

Security emerges from repeated application.



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Confusion makes the cipher hard to understand,
Diffusion makes the cipher hard to analyze statistically.

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Block Ciphers – Fiestel Network

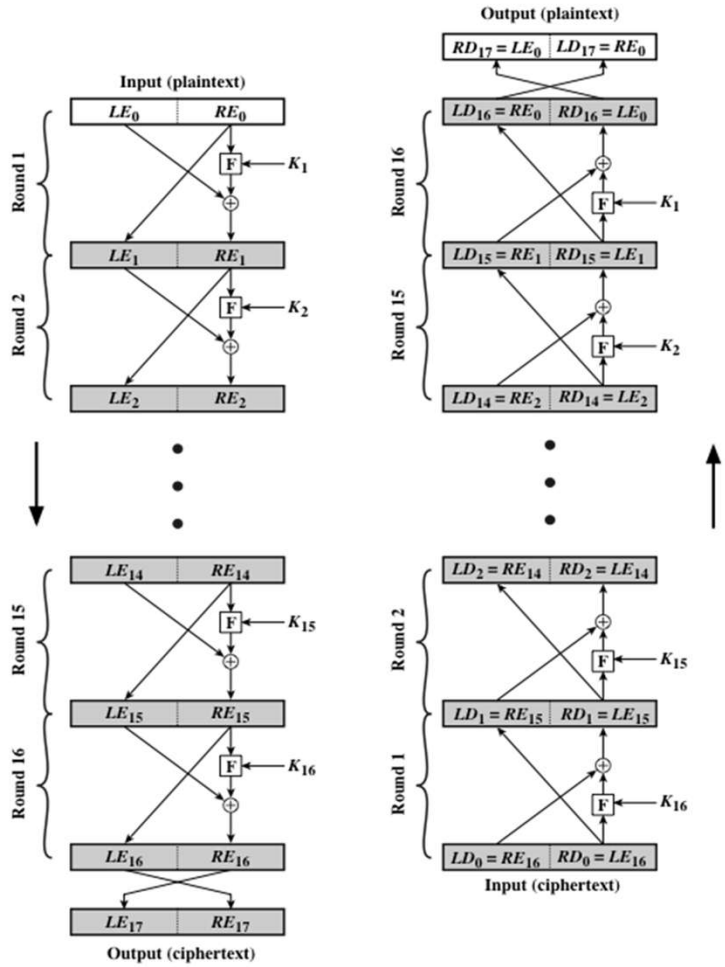


Figure 2.2 Feistel Encryption and Decryption (16 rounds)

Fiestel cipher design



Block size

- Larger block sizes mean greater security but reduced encryption/decryption speed

Key size

- Larger key size means greater security but may decrease encryption/decryption speed

Number of rounds

- The essence of a symmetric block cipher is that a single round offers inadequate security but that multiple rounds offer increasing security

Subkey generation algorithm

- Greater complexity in this algorithm should lead to greater difficulty of cryptanalysis

Round function

- Greater complexity generally means greater resistance to cryptanalysis

Fast software encryption/decryption

- In many cases, encryption is embedded in applications or utility functions in such a way as to preclude a hardware implementation; accordingly, the seed of execution of the algorithm becomes a concern

Ease of analysis

- If the algorithm can be concisely and clearly explained, it is easier to analyze that algorithm for cryptanalytic vulnerabilities and therefore develop a higher level of assurance as to its strength

Fiestel cipher

Conceptually, Split the block into L and R

Compute:

$L_{i+1} = R_i$

$R_{i+1} = L_i$ XOR $F(R_i, K_i)$

To decrypt, you can recover the previous values:

$R_i = L_{i+1}$

$L_i = R_{i+1}$ XOR $F(R_i, K_i)$

and since $R_i = L_{i+1}$, this becomes:

$L_i = R_{i+1}$ XOR $F(L_{i+1}, K_i)$

Block Cipher - DES

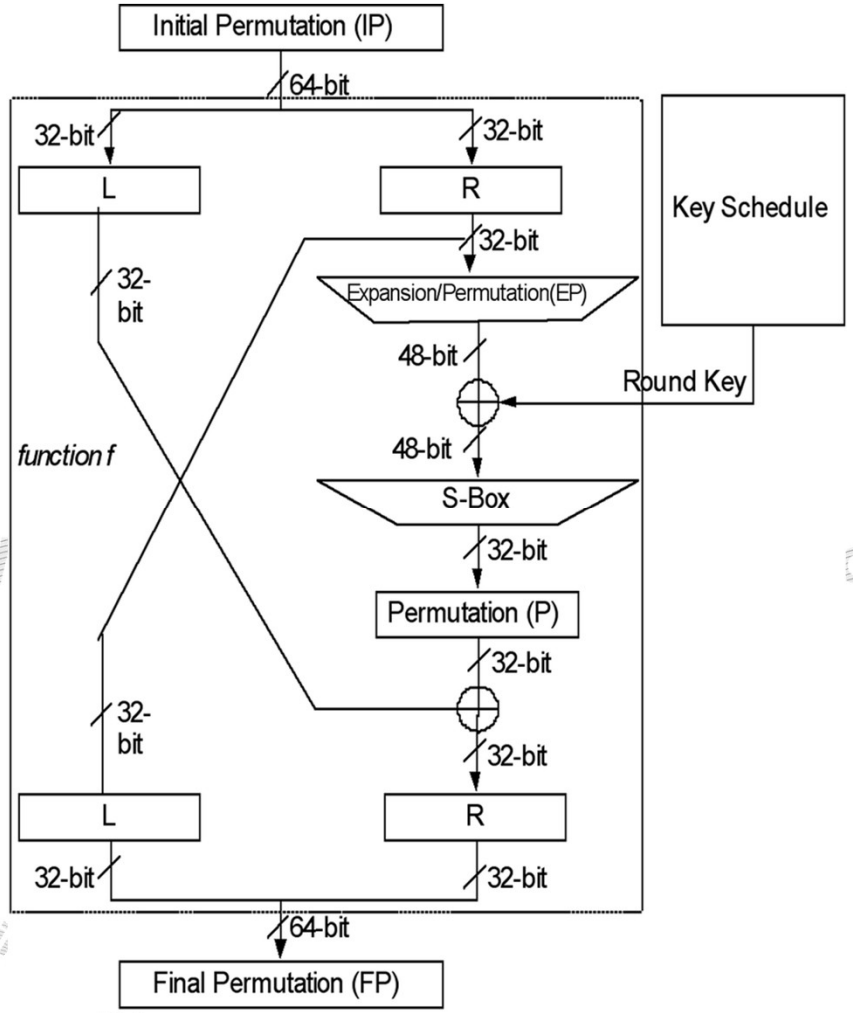
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DES has a 64-bit block length.

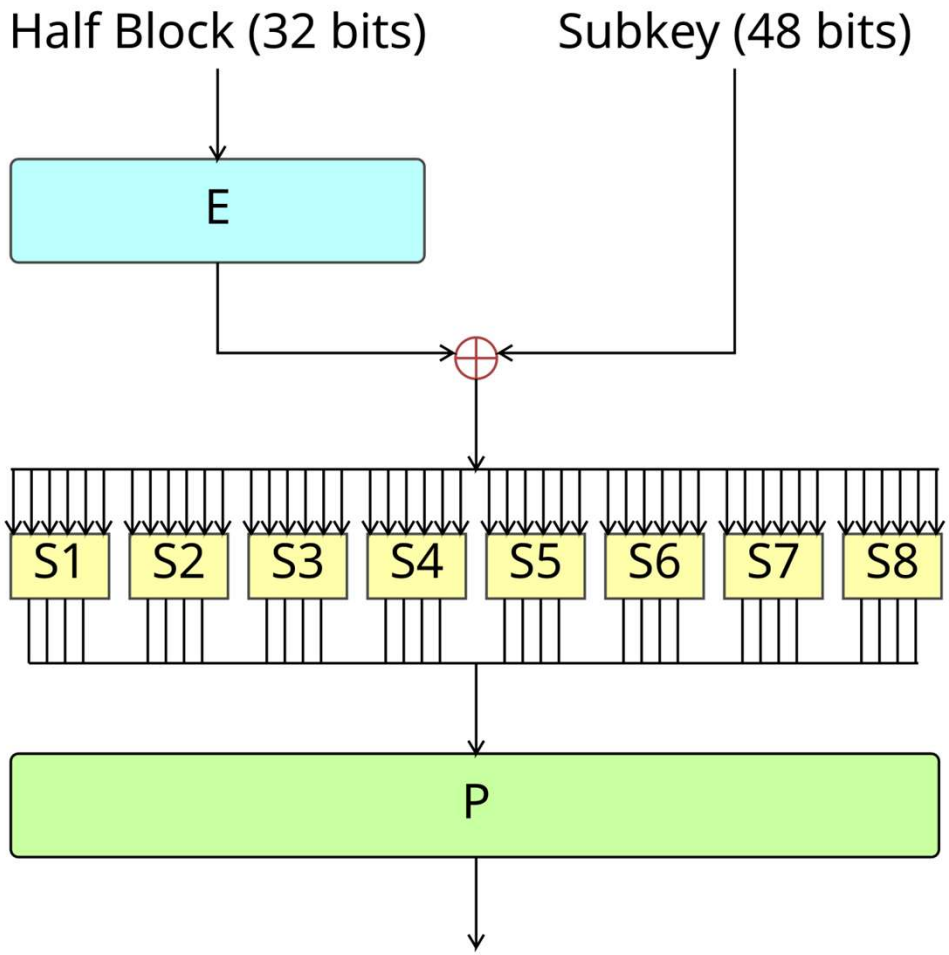
- DES uses a 56-bit key;
- Out of 16 rounds, each round of DES uses a 48-bit ***subkey*** and each **subkey** consists of a 48-bit subset of the 56-bit key.

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DES One Round



DES One Round with S boxes



DES continues

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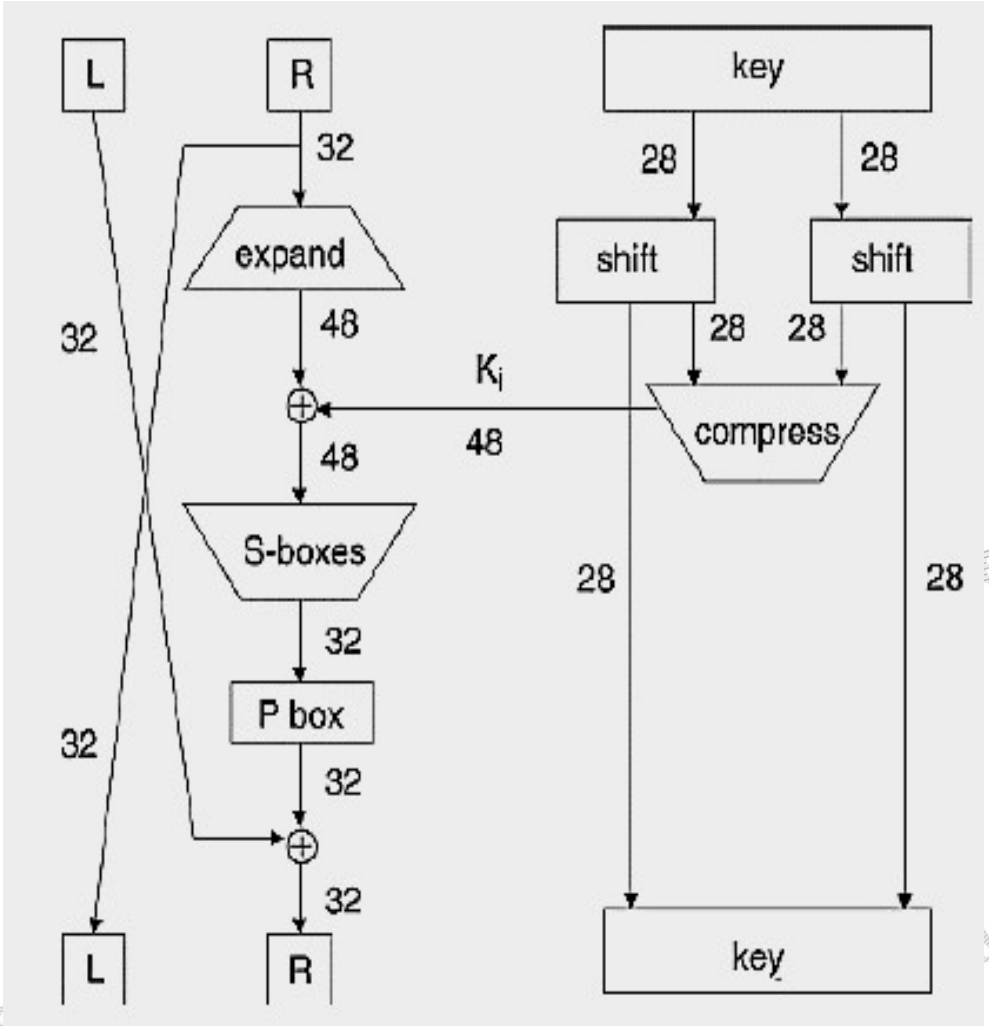
DES round function F can be written as

$$F(R_{i-1}, K_i) = \text{P-box}(\text{S-boxes}(\text{Expand}(R_{i-1}) \oplus K_i)).$$

The function F, which depends on the output of the previous round and the key K, is known as a round function

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Each round of DES



DES expansion

The 48-bit result of the DES expansion permutation consists of the bits

31	0	1	2	3	4	3	4	5	6	7	8
7	8	9	10	11	12	11	12	13	14	15	16
15	16	17	18	19	20	19	20	21	22	23	24
23	24	25	26	27	28	27	28	29	30	31	0

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where the 32-bit input is, according to our convention, numbered as

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31.

DES – S boxes

Each of the **eight** DES S-boxes maps 6 bits to 4 bits, and, consequently, each can be viewed as an array of 4 rows and 16 columns, with one "nibble" (4-bit value)

input to the S-box is denoted $b_0 b_1 b_2 b_3 b_4 b_5$

b_0b_5 \ $b_1b_2b_3b_4$	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
00	1110	0100	1101	0001	0010	1111	1011	1000	0011	1010	0110	1100	0101	1001	0000	0111
01	0000	1111	0111	0100	1110	0010	1101	0001	1010	0110	1100	1011	1001	0101	0011	1000
10	0100	0001	1110	1000	1101	0110	0010	1011	1111	1100	1001	0111	0011	1010	0101	0000
11	1111	1100	1000	0010	0100	1001	0001	0111	0101	1011	0011	1110	1010	0000	0110	1101

The permutation structure of S-box 1 is easier to see if we rewrite the bits in hexadecimal:

b_0b_5 \ $b_1b_2b_3b_4$	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	E	4	D	1	2	F	B	8	3	A	6	C	5	9	0	7
1	0	F	7	4	E	2	D	1	A	6	C	B	9	5	3	8
2	4	1	E	8	D	6	2	B	F	C	9	7	3	A	5	0
3	F	C	8	2	4	9	1	7	5	B	3	E	A	0	6	D

DES P-box

The P-box permutation is

15	6	19	20	28	11	27	16	0	14	22	25	4	17	30	9
1	7	23	13	31	26	2	8	18	12	29	5	21	10	3	24.

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DES Key Scheduling

We separate the 56 bits of the key into two group and permute them and call them Left Key (LK) group & Right Key (RK) group:

LK

49	42	35	28	21	14	7
0	50	43	36	29	22	15
8	1	51	44	37	30	23
16	9	2	52	45	38	31

RK

55	48	41	34	27	20	13
6	54	47	40	33	26	19
12	5	53	46	39	32	25
18	11	4	24	17	10	3

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DES key scheduling continued

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We define the compression mechanism as the permutation on the left part (LP) as,

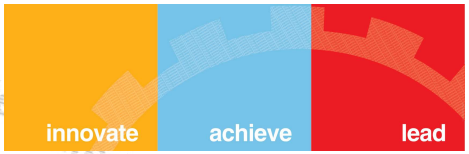
13	16	10	23	0	4	2	27	14	5	20	9
22	18	11	3	25	7	15	6	26	19	12	1

And on the right part (RP) as,

12	23	2	8	18	26	1	11	22	16	4	19
15	20	10	27	5	24	17	13	21	7	0	3

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DES key scheduling continued



Now we define the shift operation at each round r_i, for $i=1, 2, \ldots, n$

$$r_i = \begin{cases} 1 & \text{if } i \in \{1, 2, 9, 16\} \\ 2 & \text{otherwise.} \end{cases}$$

The DES key schedule algorithm for generating the 48-bit subkey K_i for, round i can now be described as

---

for each round $i = 1, 2, \ldots, n$
    LK = cyclically left shift LK by r_i bits
    RK = cyclically left shift RK by r_i bits
    The left half of subkey K_i consists of bits LP of LK
    The right half of subkey K_i consists of bits RP of RK
next i

Let us check DES key scheduling

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A commonly used example key is:
0x133457799BBCDFF1 (64-bit, includes parity)

Schedule the first key k_i

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Cryptanalysis on DES

Mathematicians are very good at solving linear equations, and the only part of DES that is not linear is the S-boxes. As a result, the S-boxes are crucial to the security of DES.

Today, DES is vulnerable simply because the key is too small, not because of any noteworthy shortcut attack.

Cryptanalysis: DES

- For DES, C and P are
- each 64 bits, while K is 56 bits. The notation that we'll adopt for the encryption of P
- with key K is
- C = E(P,K)
- while the corresponding decryption is denoted
- P = D(C,K).

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Multiple DES

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For DES, C and P are each 64 bits, while K is 56 bits. The notation that we'll adopt for the encryption of P with key K is

$$C = E(P,K)$$

while the corresponding decryption is denoted

$$P = D(C,K)$$

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"Double" DES

- DES key length is insufficient today. It turns out that there is a clever way to use DES with a larger key length. Intuitively, it seems that "double" DES might be the thing to do,

$$C = E(E(P,K1),K2).$$

- However, there is an attack on double DES that renders it more-or-less equivalent to single DES. Although the attack is somewhat **impractical**, it's close enough to being practical, that it is cause for concern. This attack is a **chosen plaintext attack (CPA)**.

Chosen Plaintext Attack on DES

- select a particular plaintext P and obtain the corresponding ciphertext C. Our goal is to find the keys K1 and K2 in equation, C =E(E(P,K1),K2).

- First we precompute a table of size 2^{56} containing the pairs *E(P,K)* and *K* for all possible key values *K*

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Meet-in-the-middle Attack

- Now given this table and the ciphertext value C corresponding to the chosen P, we decrypt C with keys K˜ until we find a value D(C, K˜) that is in table.
- The value that we find in the table will be E(P,K) for some K and we have
- D(C,K˜) = E(P,K)

where K˜ and K are known. That we have found the 112-bit key can be seen by encrypting both sides with K˜ , which gives

- C = E(E(P,K), K˜),
- that is, K1 = K and K2 = K˜

Triple DES

- It seems that the logical approach to triple DES would be
- C = E(E(E(P,K1),K2),K3)
- but this is not the way it's done. Instead, triple DES is defined as
- C = E(D(E(P,K1),K2),K1).
- Notice that triple DES only uses two keys, and encrypt-decrypt-encrypt, or EDE

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